ICCT COLLEGES FOUNDATION INC.
V.V Soliven Avenue II Cainta Rizal
College of Engineering and Digital Technology

LOCAL AREA NETWORK DESIGN

A Project

Presented to the

Faculty of College of Engineering and Digital Technology

ICCT Colleges Foundation Inc.

V.V. Soliven Avenue II Cainta Rizal

Bachelor of Science Major in Computer Engineering

BY.

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CHAPTER 6: SAFETY PRECAUTIONS

6.0 SAFETY PRECAUTIONS

These rules must be observed before, during and after all work to ensure a safe and stable working environment. Having a proper Guidelines helps prevent accidents, protects workers from injury and keep the entire network running reliably.

1. All local electrical safety regulations, building codes, and industry standards must be followed at all times.
2. All plugs, cables, and wiring must be properly secured and maintained. Any damaged, exposed, or loose wires must be replaced right away to avoid hazards.
3. Each piece of equipment must be used only within its stated specifications and intended purpose to prevent overheating, short circuits, or hardware damage.
4. Work areas and equipment rooms must be kept clean, dry, and well-ventilated to reduce risks from moisture, dust, and overheating.
5. Only trained and authorized personnel should perform installation, repair, or maintenance tasks to ensure correct handling and avoid mistakes.
6. All safety measures must be followed consistently to protect both people and the network infrastructure from harm or failure.

6.1 ELECTRICAL SAFETY

All network devices run on electricity, so following these rules prevents shocks, overheating, and fire risks:

1. Always check that the power source matches the voltage and current requirements of each device — such as Dell desktops, D-Link switches, and TP-Link routers — to avoid damage or overheating.
2. Do not plug too many devices into one outlet or power strip, as this causes overload and raises the risk of short circuits.
3. Make sure all plugs and connections fit tightly; loose connections create heat and can lead to sparks or power failure.
4. Never touch electrical parts or plugs with wet hands, and keep all equipment away from water or damp areas.
5. Turn off and unplug all devices before installing, moving, or repairing any wiring or hardware.
6. During storms or lightning, disconnect sensitive equipment from power and network lines to avoid damage from sudden voltage surges.

7. se only certified, high-quality cables and power supplies. Low-quality materials can fail easily and create hazards.
8. Never look directly into a fiber optic cable, as the invisible light can harm your eyes.

6.2 POWER CONTACT INDUCTION

When electricity flows through wires, it creates magnetic fields that can affect nearby cables. Follow these guidelines:

1. Keep power cables separate from data cables (CAT6 or fiber) by at least 30 cm, or cross them only at 90 degrees, to avoid signal interference and unwanted voltage buildup.
2. Work involving high-power circuits or transformers must be done only by trained and authorized personnel.
3. Never modify, bypass, or open power supply units or electrical coils, as this removes built-in safety protections.
4. Check power units regularly for signs of overheating, unusual smells, or damage; replace any faulty parts immediately.
5. Ensure good ventilation around power equipment so it stays within safe operating temperatures.

6.3 AC AND DC SAFETY

The network uses both AC power from the building's main supply and lower-voltage DC power inside devices:

1. Follow local electrical codes and standards for wiring and power distribution.
2. When testing with tools like a multimeter, select the correct setting for AC or DC to avoid wrong readings or damage.
3. Use cables rated for the amount of power they carry; underrated wires can overheat and cause faults.
4. Always use the original or recommended power adapters for each device. Mismatched adapters can destroy hardware or cause fires.
5. Wear protective gear such as insulated gloves and rubber-soled shoes when working on electrical connections.
6. Confirm all power is off before starting work, and double-check all connections before turning it back on.

6.4 GROUNDING SAFETY

Grounding provides a safe path for excess electricity to flow into the earth, protecting both people and devices:

1. All metal racks, switches, servers, routers, and other equipment must be properly grounded according to electrical standards.
2. Never remove or cut off the grounding pin on plugs or wires — it is your main protection against shock.
3. Grounding installation and testing should only be done by qualified technicians, as incorrect grounding is useless and dangerous.
4. Inspect grounding connections regularly to ensure they remain tight, clean, and free from rust or damage.
5. Always disconnect power before working on grounding wires.

6.5 ELECTRIC SHOCK PREVENTION

Electric shock happens when current passes through the body. These steps reduce this risk:

1. Never insert tools or fingers into sockets or open ports.
2. Wear proper protective equipment when handling wiring or power sources.
3. Ensure every circuit has a fuse or circuit breaker that automatically cuts power during overloads or faults.
4. Replace any cables that are frayed, cracked, or have exposed wires immediately.
5. When unplugging devices, pull the plug itself, not the cable, to avoid breaking internal wires.
6. Keep the work area dry and use insulated mats when working near electrical parts.
7. If someone receives a shock, turn off the power first before helping them.

6.6 EQUIPMENT PROTECTION

Taking care of hardware and software ensures longer life and fewer problems:

1. Keep the server room and equipment areas well-ventilated and at a stable temperature (between 10°C and 35°C) to prevent overheating.
2. Install Uninterruptible Power Supplies (UPS) and surge protectors to guard against power outages and sudden spikes.
3. Organize cables neatly and avoid bending them too sharply, especially fiber cables, which can break if bent too much.

4. Limit access to the server room and equipment racks to only authorized staff to prevent tampering or theft.
5. Use strong passwords, firewalls, and antivirus software to protect both the network and stored data.
6. Shut down devices properly using the correct procedure to avoid data loss or internal damage.
7. Perform regular cleaning and inspections to remove dust and check for loose connections or signs of wear.